

Lecture-09 [25 August 2026]

Span and Span Contractions

Definition (Span of a vector)

Given $\underline{v} \in \mathbb{R}^{|\mathcal{S}|}$, the span of $\underline{v}$, denoted $sp(\underline{v})$, is defined as

$$sp(\underline{v}) = \max_{s \in \mathcal{S}} v(s) - \min_{s \in \mathcal{S}} v(s).$$

Exercise: Show that span is a semi-norm:

$$(1) \quad sp(\underline{v}) \geq 0 \quad \forall \underline{v} \in \mathbb{R}^{|\mathcal{S}|}.$$

$$(2) \quad sp(\underline{v} + \underline{u}) \leq sp(\underline{v}) + sp(\underline{u}) \quad \forall \underline{u}, \underline{v} \in \mathbb{R}^{|\mathcal{S}|}.$$

$$(3) \quad sp(\alpha \underline{v}) = |\alpha| sp(\underline{v}) \quad \forall \alpha \in \mathbb{R}, \forall \underline{v} \in \mathbb{R}^{|\mathcal{S}|}.$$

Furthermore, show that

$$(4) \quad sp(\underline{v} + \alpha \underline{1}) = sp(\underline{v}) \quad \forall \alpha \in \mathbb{R}, \forall \underline{v} \in \mathbb{R}^{|\mathcal{S}|}.$$

$$(5) \quad sp(\underline{v}) = sp(-\underline{v}) \quad \forall \underline{v} \in \mathbb{R}^{|\mathcal{S}|}.$$

$$(6) \quad sp(\underline{v}) \leq 2 \|\underline{v}\|_{\infty}.$$

Key observation: The closer the $sp(\underline{v})$ is to zero, the closer $\underline{v}$ is to a constant vector.

Proposition 5.3:

Let Q be a $K \times M$ row-stochastic matrix in which every row is a PMF on $\{1, \dots, M\}$. Then, for any $\underline{v} \in \mathbb{R}^M$,

$$sp(Q\underline{v}) \leq \gamma sp(\underline{v}),$$

where

$$\gamma = 1 - \min_{k, k'} \sum_{m=1}^M \min\{Q(m|k), Q(m|k')\}.$$

Furthermore, there exists $\underline{v}^* \in \mathbb{R}^M$ for which $sp(Q\underline{v}^*) = \gamma sp(\underline{v}^*)$.

$$P_\phi(s'|s) = \sum_{a \in \mathcal{A}} \phi(a|s) P(s'|s, a).$$

The constant γ is referred to as the **delta-coefficient** of Q .

Applying Proposition 5.3 to $Q = P_\phi$ for any $\phi: \mathcal{S} \rightarrow \Delta(\mathcal{A})$ gives the following result.

Proposition 5.4:

For any $\phi: \mathcal{S} \rightarrow \Delta(\mathcal{A})$,

$$sp(P_\phi \underline{v}) \leq \gamma_\phi sp(\underline{v}) \quad \text{for all } \underline{v} \in \mathbb{R}^{|\mathcal{S}|},$$

where

$$\gamma_\phi := 1 - \min_{s, s' \in \mathcal{S}} \sum_{s'' \in \mathcal{S}} \min \{ P_\phi(s''|s), P_\phi(s''|s') \}.$$

Moreover, there exists $\underline{v} \in \mathbb{R}^{|\mathcal{S}|}$ for which $sp(P_\phi \underline{v}) = \gamma_\phi sp(\underline{v})$.

Here, γ_ϕ is called the delta coefficient of P_ϕ .

Remarks:

① An alternative expression for the delta-coefficient is given by

$$\begin{aligned} \gamma &= \max_{k, k'} \frac{1}{2} \sum_{m=1}^M |Q(m|k) - Q(m|k')| \\ &= \max_{k, k'} TV(Q(\cdot|k), Q(\cdot|k')), \end{aligned}$$

where TV denotes the total variation distance.

② If $P_\phi = I$ or a column permutation of I , then $\gamma_\phi = 1$.

If all rows of P_ϕ are identical, then $\gamma_\phi = 0$.

③ Notice that for any $\phi: \mathcal{S} \rightarrow \Delta(\mathcal{A})$,

$$\|P_\phi \underline{v}\|_\infty \leq \|P_\phi\| \cdot \|\underline{v}\|_\infty = \|\underline{v}\|_\infty,$$

$$\text{while } sp(P_\phi \underline{v}) \leq \gamma_\phi sp(\underline{v}).$$

Thus, P_ϕ could be a contraction under $sp(\cdot)$, but not under $\|\cdot\|_\infty$.

Proposition 5.5 [Sufficient conditions for $\gamma_\phi < 1$]

Fix $\phi: \mathcal{S} \rightarrow \Delta(A)$ and consider the stationary policy $\pi^\phi = \{\phi, \phi, \dots\}$. Recall that under π^ϕ , the state process $\{S_n\}_{n \geq 0}$ is a time-homogeneous DTMC.

- ① If $\{S_n\}_{n \geq 0}$ is irreducible and aperiodic, then $\gamma_\phi < 1$.
- ② If $\{S_n\}_{n \geq 0}$ has a single set of recurrent states and a non-empty set of transient states, then $\gamma_\phi < 1$.

Theorem 5.11 : [L is a contraction under sp]

Fix $\lambda \in [0, 1)$ and consider the operator $L: \mathbb{R}^{|\mathcal{S}|} \rightarrow \mathbb{R}^{|\mathcal{S}|}$ defined by

$$L\underline{v}(s) = \max_{a \in A} \left[r(s, a) + \lambda \sum_{s' \in \mathcal{S}} P(s'|s, a) v(s') \right], \quad s \in \mathcal{S}.$$

Then, there exists $\gamma^* \in [0, 1]$ such that

$$sp(L\underline{v} - L\underline{u}) \leq \lambda \gamma^* sp(\underline{v} - \underline{u}) \quad \forall \underline{u}, \underline{v} \in \mathbb{R}^{|\mathcal{S}|}.$$

Proof of Theorem 5.11

Fix arbitrary $\underline{u}, \underline{v} \in \mathbb{R}^{|\mathcal{S}|}$. Let

$$\phi_{\underline{v}} := \arg \max_{\phi} L_{\phi} \underline{v}, \quad \phi_{\underline{u}} = \arg \max_{\phi} L_{\phi} \underline{u}.$$

Then, we have

$$\begin{aligned} L\underline{v} - L\underline{u} &= L_{\phi_{\underline{v}}} \underline{v} - L\underline{u} \\ &\leq L_{\phi_{\underline{v}}} \underline{v} - L_{\phi_{\underline{v}}} \underline{u} \\ &= r_{\phi_{\underline{v}}} + \lambda P_{\phi_{\underline{v}}} \underline{v} - \left(r_{\phi_{\underline{v}}} + \lambda P_{\phi_{\underline{v}}} \underline{u} \right) \\ &= \lambda P_{\phi_{\underline{v}}} (\underline{v} - \underline{u}). \end{aligned}$$

$\phi_{\underline{v}}, \phi_{\underline{u}}: \mathcal{S} \rightarrow A$ are

stationary, deterministic
decision rules

Similarly,

$$L\underline{v} - L\underline{u} = L\underline{v} - L_{\phi_{\underline{u}}} \underline{u}$$

$$\geq L_{\phi_{\underline{u}}} \underline{v} - L_{\phi_{\underline{u}}} \underline{u}$$

$$= \lambda P_{\phi_{\underline{u}}}(\underline{v} - \underline{u}).$$

It then follows that

$$\max_{s \in \mathcal{S}} L_{\underline{v}}(s) - L_{\underline{u}}(s) \leq \lambda \max_{s \in \mathcal{S}} P_{\phi_{\underline{v}}}(\underline{v} - \underline{u})(s)$$

$$= \lambda \max_{s \in \mathcal{S}} \sum_{s' \in \mathcal{S}} P(s'|s, \phi_{\underline{v}}(s)) (v(s') - u(s')).$$

Similarly,

$$\min_{s \in \mathcal{S}} L_{\underline{v}}(s) - L_{\underline{u}}(s) \geq \lambda \min_{s \in \mathcal{S}} P_{\phi_{\underline{u}}}(\underline{v} - \underline{u})(s)$$

$$= \lambda \min_{s \in \mathcal{S}} \sum_{s' \in \mathcal{S}} P(s'|s, \phi_{\underline{u}}(s)) (v(s') - u(s')).$$

Proposition 5.6: $P(\cdot | s, a) \rightarrow$ PMF on $\mathcal{S}$

Let $P = \{P(s'|s, a) : (s, a, s') \in \mathcal{S} \times \mathcal{A} \times \mathcal{S}\}$ denote the TPM of an MDP.

For any $\underline{v} \in \mathbb{R}^{|\mathcal{S}|}$, define

$$sp(P\underline{v}) = \left\{ \max_{(s, a) \in \mathcal{S} \times \mathcal{A}} \sum_{s' \in \mathcal{S}} P(s'|s, a) v(s') \right\} - \left\{ \min_{(s, a) \in \mathcal{S} \times \mathcal{A}} \sum_{s' \in \mathcal{S}} P(s'|s, a) v(s') \right\}.$$

Then,

$$sp(P\underline{v}) \leq \gamma sp(\underline{v}),$$

where

$$\gamma := 1 - \min_{\substack{(s, a) \\ (s', a')}} \sum_{s'' \in \mathcal{S}} \min \{P(s''|s, a), P(s''|s', a')\}.$$

Applying Proposition 5.6, we get

$$sp(L_{\underline{v}} - L_{\underline{u}}) \leq \lambda \max_{s \in \mathcal{S}} \sum_{s' \in \mathcal{S}} P(s'|s, \phi_{\underline{v}}(s)) (v(s') - u(s'))$$

$$- \lambda \min_{s \in \mathcal{S}} \sum_{s' \in \mathcal{S}} P(s'|s, \phi_{\underline{u}}(s)) (v(s') - u(s'))$$

$$\leq \lambda \max_{(s,a) \in \mathcal{S} \times \mathcal{A}} \sum_{s' \in \mathcal{S}} P(s'|s,a) (v(s') - u(s'))$$

$$- \lambda \min_{(s,a) \in \mathcal{S} \times \mathcal{A}} \sum_{s' \in \mathcal{S}} P(s'|s,a) (v(s') - u(s'))$$

$$\leq \lambda \operatorname{sp}(P(v-u))$$

$$\stackrel{\text{Prop. 5.6}}{\leq} \lambda \gamma^* \operatorname{sp}(v-u).$$

Remark :

Because $\lambda \gamma^* \leq \lambda$, algorithms that start with an initial $\underline{v}_0 \in \mathbb{R}^{|\mathcal{S}|}$ and recursively apply

$$\underline{v}_k \leftarrow L \underline{v}_{k-1}$$

are likely to stop faster when using "sp" as stopping criterion instead of $\|\cdot\|_\infty$.

Corollary 5.4 [Corollary to Theorem 5.11]

For any stationary and deterministic $\phi: \mathcal{S} \rightarrow \mathcal{A}$,

$$\operatorname{sp}(L_\phi \underline{v} - L_\phi \underline{u}) \leq \lambda \gamma_\phi \operatorname{sp}(\underline{v} - \underline{u}),$$

where

$$\gamma_\phi = 1 - \min_{s,s'} \sum_{s'' \in \mathcal{S}} \min \{ P_\phi(s''|s), P_\phi(s''|s') \}.$$

Properties of $L \underline{v} - \underline{v}$

Define the optimal one-step improvement operator $B: \mathbb{R}^{|\mathcal{S}|} \rightarrow \mathbb{R}^{|\mathcal{S}|}$ as

$$B \underline{v} = L \underline{v} - \underline{v}, \quad \underline{v} \in \mathbb{R}^{|\mathcal{S}|}.$$

The Bellman equation $L \underline{v} = \underline{v}$ translates to

$$B \underline{v} = \underline{0}.$$



Bellman optimality equation in terms of B

$$f(y) \geq f(x) + f'(x)(y-x) \quad \forall x, y$$

Lemma 5.5: [Convexity-like Property of B and L Operators]

For all $\underline{u}, \underline{v} \in \mathbb{R}^{|S|}$, the following component-wise relations hold:

$$B\underline{v} \geq B\underline{u} + (\lambda P_{\phi_{\underline{u}}} - \mathbb{I})(\underline{v} - \underline{u}),$$

f convex:

$$L\underline{v} - \underline{v} \geq L\underline{u} - \underline{u} + (\lambda P_{\phi_{\underline{u}}} - \mathbb{I})(\underline{v} - \underline{u}),$$

$$f(y) \geq f(x) + f'(x)(y-x)$$

where

$$\phi_{\underline{u}} = \arg \max_{\phi} L_{\phi} \underline{u}$$

denotes the optimal stationary deterministic (greedy) decision rule for $\underline{u}$.

Proof of Lemma 5.5

We have

$$\begin{aligned} L\underline{v} &= \sup_{\phi} L_{\phi} \underline{v} \\ &\geq L_{\phi_{\underline{u}}} \underline{v} \\ &= r_{\phi_{\underline{u}}} + \lambda P_{\phi_{\underline{u}}} \underline{v} \\ &= r_{\phi_{\underline{u}}} + \lambda P_{\phi_{\underline{u}}} \underline{u} + \lambda P_{\phi_{\underline{u}}} (\underline{v} - \underline{u}) \\ &= L_{\phi_{\underline{u}}} \underline{u} + \lambda P_{\phi_{\underline{u}}} (\underline{v} - \underline{u}) \\ &= \sup_{\phi} L_{\phi} \underline{u} + \lambda P_{\phi_{\underline{u}}} (\underline{v} - \underline{u}) \\ &= L\underline{u} + \lambda P_{\phi_{\underline{u}}} (\underline{v} - \underline{u}). \end{aligned}$$

Subtracting $\underline{v} - \underline{u}$ from both sides, we get the desired result.

Some notations going forward

$$\min(\underline{v}) = \min_{s \in S} v(s), \quad \max(\underline{v}) = \max_{s \in S} v(s).$$

Theorem 5.12

Fix an arbitrary $\underline{v} \in \mathbb{R}^{|\mathcal{S}|}$. Let $\phi_{\underline{v}} = \arg \max_{\phi} L_{\phi} \underline{v}$. Then:

$$\begin{aligned} \underline{v} + \frac{1}{1-\lambda} \max(L_{\underline{v}} - \underline{v}) \underline{1} &\stackrel{(1)}{\geq} L_{\underline{v}} + \frac{\lambda}{1-\lambda} \max(L_{\underline{v}} - \underline{v}) \underline{1} \stackrel{(2)}{\geq} v_{\infty, \lambda}^* \stackrel{(3)}{\geq} v_{\infty, \lambda}^{\phi_{\underline{v}}} \\ &\stackrel{(4)}{\geq} L_{\underline{v}} + \frac{\lambda}{1-\lambda} \min(L_{\underline{v}} - \underline{v}) \underline{1} \stackrel{(5)}{\geq} \underline{v} + \frac{1}{1-\lambda} \min(L_{\underline{v}} - \underline{v}) \underline{1}. \end{aligned}$$

Proof of Theorem 5.12

(3) follows straightforwardly. We will derive (4) and (5) below.

For deriving (1) and (2), one may simply reverse the roles of $\underline{v}$ and $v_{\infty, \lambda}^*$ in the derivation of (4), (5).

Applying Lemma 5.5 with $\underline{v} = v_{\infty, \lambda}^*$ and $\underline{u} = \underline{v}$ gives

$$\underbrace{L_{v_{\infty, \lambda}^*} - v_{\infty, \lambda}^*}_{= \underline{0}} \geq L_{\underline{v}} - \underline{v} + (\lambda P_{\phi_{\underline{v}}} - \mathbf{I})(v_{\infty, \lambda}^* - \underline{v})$$

Applying $(\mathbf{I} - \lambda P_{\phi_{\underline{v}}})^{-1}$ to both sides, and recalling (from lecture-07) that

$$\forall \phi, \quad (\mathbf{I} - \lambda P_{\phi})^{-1} \underline{u} \geq (\mathbf{I} - \lambda P_{\phi})^{-1} \underline{v} \quad \text{whenever } \underline{u} \geq \underline{v},$$

We get

$$\begin{aligned} \underline{0} &\geq (\mathbf{I} - \lambda P_{\phi_{\underline{v}}})^{-1} (L_{\underline{v}} - \underline{v}) - (v_{\infty, \lambda}^* - \underline{v}) \\ \Leftrightarrow v_{\infty, \lambda}^* &\geq \boxed{\underline{v} + (\mathbf{I} - \lambda P_{\phi_{\underline{v}}})^{-1} (L_{\underline{v}} - \underline{v})} \xrightarrow{\phi_{\underline{v}}} v_{\infty, \lambda}^{\phi_{\underline{v}}} \\ &= \underline{v} + \left(\mathbf{I} + \lambda P_{\phi_{\underline{v}}} + \lambda^2 P_{\phi_{\underline{v}}}^2 + \dots \right) (L_{\underline{v}} - \underline{v}) \\ &= \underline{v} + L_{\underline{v}} - \underline{v} + \lambda P_{\phi_{\underline{v}}} \left(\mathbf{I} + \lambda P_{\phi_{\underline{v}}} + \lambda^2 P_{\phi_{\underline{v}}}^2 + \dots \right) (L_{\underline{v}} - \underline{v}) \\ &\geq \underline{v} + L_{\underline{v}} - \underline{v} + \min(L_{\underline{v}} - \underline{v}) \cdot \lambda P_{\phi_{\underline{v}}} \left(\mathbf{I} + \lambda P_{\phi_{\underline{v}}} + \lambda^2 P_{\phi_{\underline{v}}}^2 + \dots \right) \underline{1} \\ &= \underline{v} + L_{\underline{v}} - \underline{v} + \frac{\lambda}{1-\lambda} \min(L_{\underline{v}} - \underline{v}) \cdot \underline{1} \end{aligned}$$

$$\begin{aligned}
&\geq \underline{v} + \min(L\underline{v} - \underline{v}) \underline{1} + \frac{\lambda}{1-\lambda} \min(L\underline{v} - \underline{v}) \underline{1} \\
&= \underline{v} + \frac{1}{1-\lambda} \cdot \min(L\underline{v} - \underline{v}) \underline{1}.
\end{aligned}$$

Stopping Criteria and Value Iteration

The following result will be very useful when analysing the popular value iteration algorithm later.

Theorem 5.13

Fix $\underline{v} \in \mathbb{R}^{|\mathcal{S}|}$ and $\varepsilon > 0$. If

$$\text{sp}(L\underline{v} - \underline{v}) < \left(\frac{1-\lambda}{\lambda}\right) \varepsilon,$$

then

$$\left\| L\underline{v} + \frac{\lambda}{1-\lambda} \min(L\underline{v} - \underline{v}) \underline{1} - v_{\infty, \lambda}^* \right\|_{\infty} < \varepsilon.$$

Furthermore, if $\phi_{\underline{v}} = \arg\max_{\phi} L_{\phi} \underline{v}$, then

$$\left\| v_{\infty, \lambda}^{\phi_{\underline{v}}} - v_{\infty, \lambda}^* \right\|_{\infty} < \varepsilon.$$

Proof: For any vectors $\underline{x}, \underline{y}, \underline{z}, \underline{w} \in \mathbb{R}^{|\mathcal{S}|}$ such that

$$\underline{x} \geq \underline{y} \geq \underline{z} \geq \underline{w},$$

we have

$$\underline{x} - \underline{w} \geq \underline{y} - \underline{z} \geq \underline{0},$$

$$\underline{x} - \underline{w} \geq \underline{y} - \underline{w} \geq \underline{0},$$

and hence

$$\|\underline{x} - \underline{w}\|_{\infty} \geq \|\underline{y} - \underline{z}\|_{\infty},$$

$$\|\underline{x} - \underline{w}\|_{\infty} \geq \|\underline{y} - \underline{w}\|_{\infty}.$$

The result follows by applying Theorem 5.12 with

$$\underline{x} = \underline{L}\underline{v} + \frac{\lambda}{1-\lambda} \max(\underline{L}\underline{v} - \underline{v}) \underline{1},$$

$$\underline{y} = v_{\infty, \lambda}^*,$$

$$\underline{z} = v_{\infty, \lambda}^{\phi_v},$$

$$\underline{w} = \underline{L}\underline{v} + \frac{\lambda}{1-\lambda} \min(\underline{L}\underline{v} - \underline{v}) \underline{1},$$

and noting that

$$\begin{aligned} \text{sp}(\underline{L}\underline{v} - \underline{v}) &= \max(\underline{L}\underline{v} - \underline{v}) - \min(\underline{L}\underline{v} - \underline{v}), \\ \|\underline{1}\|_{\infty} &= 1. \end{aligned}$$

Value Iteration Algorithm

- Input $\lambda \in [0, 1)$, $\varepsilon > 0$, $\sigma \geq \left(\frac{1-\lambda}{\lambda}\right)\varepsilon$.

- Initialise $\underline{v}' = \{v'(s) : s \in \mathcal{S}\}$

- While $\sigma \geq \left(\frac{1-\lambda}{\lambda}\right)\varepsilon$:

• $\underline{v} \leftarrow \underline{v}'$.

• Set $\underline{v}' \leftarrow \underline{L}\underline{v}$ $v'(s) = \max_a \left[r(s, a) + \sum_{s' \in \mathcal{S}} P(s'|s, a) v(s') \right]$

• $\sigma \leftarrow \text{sp}(\underline{v}' - \underline{v}) = \text{sp}(\underline{L}\underline{v} - \underline{v})$.

- Terminate and return $\phi_{\varepsilon} : \mathcal{S} \rightarrow \mathcal{A}$ and $v_{\infty, \lambda}^{\varepsilon} : \mathcal{S} \rightarrow \mathbb{R}$ as :

$$\phi_{\varepsilon}(s) = \arg \max_{a \in \mathcal{A}} \left[r(s, a) + \lambda \sum_{s' \in \mathcal{S}} P(s'|s, a) v(s') \right], \quad s \in \mathcal{S}$$

$$\begin{aligned} v_{\infty, \lambda}^{\varepsilon} &= \underline{v}' + \frac{\lambda}{1-\lambda} \min(\underline{v}' - \underline{v}) \underline{1} \\ &= \underline{L}\underline{v} + \frac{\lambda}{1-\lambda} \min(\underline{L}\underline{v} - \underline{v}) \underline{1}. \end{aligned}$$

$$\begin{aligned}
\underline{v} + (\mathbf{I} - \lambda P_{\phi_{\underline{v}}})^{-1} (L\underline{v} - \underline{v}) &= \underline{v} + (\mathbf{I} - \lambda P_{\phi_{\underline{v}}})^{-1} (L_{\phi_{\underline{v}}} \underline{v} - \underline{v}) \\
&= \underline{v} + (\mathbf{I} - \lambda P_{\phi_{\underline{v}}})^{-1} \begin{bmatrix} x_{\phi_{\underline{v}}} + \lambda P_{\phi_{\underline{v}}} \underline{v} & -\underline{v} \end{bmatrix} \\
&= \underline{v} + (\mathbf{I} - \lambda P_{\phi_{\underline{v}}})^{-1} \begin{bmatrix} x_{\phi_{\underline{v}}} - (\mathbf{I} - \lambda P_{\phi_{\underline{v}}}) \underline{v} \end{bmatrix} \\
&= (\mathbf{I} - \lambda P_{\phi_{\underline{v}}})^{-1} x_{\phi_{\underline{v}}}
\end{aligned}$$

$$= \frac{v^{\phi_{\underline{v}}}}{\omega_r \lambda}.$$